

Paper 1 - Set 04 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Quadratic functions
2	B	Differentiation
3	D	Number, units and estimation
4	C	Integration
5	F	Sequences and series
6	C	GCSE algebra
7	E	Pythagoras, mensuration and 3D
8	B	Straight lines
9	D	Binomial expansion
10	B	Binomial expansion
11	B	Functions and transformations
12	D	Number, units and estimation
13	E	Graphs of functions
14	C	Differentiation
15	D	Triangles, radians and sectors
16	F	Polynomials, factor and remainder theorems
17	A	Sequences and series
18	D	Quadratic functions
19	F	Indices and surds
20	D	Triangles, radians and sectors

Worked solutions

Question 1 QUA-18 · Quadratic functions · spec MM1.3

Find the range of values of x for which $x^2 - 7x + 10 < 0$ and $x^2 - 4 > 0$.

Answer: A

$(x - 2)(x - 5) < 0$ gives $2 < x < 5$; $x^2 > 4$ gives $x < -2$ or $x > 2$. The intersection is $2 < x < 5$.

Question 2 DIF-04 · Differentiation · spec MM6.3

Find the equation of the normal to the curve $y = \sqrt{x}$ at the point $(4, 2)$.

Answer: B

$y' = \frac{1}{2\sqrt{x}} = \frac{1}{4}$ at $x = 4$, so the normal has gradient -4 : $y - 2 = -4(x - 4)$, i.e. $y = -4x + 18$.

Question 3 NUM-01 · Number, units and estimation · spec M2.3

Find the highest common factor of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$.

Answer: D

Take the lower power of each common prime: $2^2 \times 3^2 = 36$.

Question 4 INT-17 · Integration · spec MM7.6

The gradient of a curve at (x, y) is $3\sqrt{x}$ and the curve passes through $(4, 20)$. Find the y-intercept of the curve.

Answer: C

$y = 2x^{3/2} + c$; $20 = 16 + c \Rightarrow c = 4$. The y-intercept is 4.

Question 5 SEQ-15 · Sequences and series · spec MM2.3

Evaluate $\sum_{k=1}^{10} 3 \times 2^k$.

Answer: F

This is $6 + 12 + \dots + 6 \cdot 2^9$: first term 6, ratio 2, 10 terms: $6(2^{10} - 1) = 6 \times 1023 = 6138$.

Question 6 ALG-15 · GCSE algebra · spec M4.18

The distance-time graph of a journey is a straight line from $(0, 0)$ to $(2, 60)$, then horizontal to $(3, 60)$, then a straight line to $(5, 0)$ (time in hours, distance from home in km). What was the average speed for the whole journey?

Answer: C

Total distance $60 + 60 = 120$ km in 5 hours: 24 km/h. (Displacement is 0 but distance travelled is not.)

Question 7 MEN-07 · Pythagoras, mensuration and 3D · spec M5.7

A ladder of length 6.5 m leans against a vertical wall with its foot 2.5 m from the wall on horizontal ground. How far up the wall does it reach?

Answer: E

$\sqrt{6.5^2 - 2.5^2} = \sqrt{42.25 - 6.25} = \sqrt{36} = 6$.

Question 8 LIN-07 · Straight lines · spec MM3.1

The line $ax + by = 12$ has x -intercept 3 and y -intercept -4 . Find $a - b$.

Answer: B

$$(3, 0): 3a = 12 \Rightarrow a = 4. (0, -4): -4b = 12 \Rightarrow b = -3. a - b = 7.$$

Question 9 BIN-06 · Binomial expansion · spec MM2.4

In the expansion of $(1 + x)^n$, the coefficients of x^4 and x^5 are equal. Find n .

Answer: D

$$\binom{n}{4} = \binom{n}{5} \text{ requires } 4 + 5 = n \text{ (by symmetry } \binom{n}{r} = \binom{n}{n-r} \text{)}, \text{ so } n = 9.$$

Question 10 BIN-01 · Binomial expansion · spec MM2.4

Find the coefficient of x^3 in the expansion of $(2 + 3x)^5$.

Answer: B

$$\binom{5}{3} 2^2 (3x)^3 = 10 \times 4 \times 27x^3 = 1080x^3.$$

Question 11 FUN-17 · Functions and transformations · spec MM8.3

The line $y = mx + c$ passes through $(2, 7)$. If m is increased by 1 and c is decreased by 2, the new line passes through $(2, k)$. Find k .

Answer: B

$$\text{New value at } x = 2: (m + 1) \cdot 2 + (c - 2) = 2m + c = 7. \text{ So } k = 7.$$

Question 12 NUM-11 · Number, units and estimation · spec M2.2

Evaluate $\left(2\frac{1}{3} - 1\frac{3}{4}\right) \div \frac{7}{6}$.

Answer: D

$$\frac{7}{3} - \frac{7}{4} = \frac{28-21}{12} = \frac{7}{12}; \frac{7}{12} \times \frac{6}{7} = \frac{1}{2}.$$

Question 13 GRA-16 · Graphs of functions · spec MM8.7

The line $y = mx$ meets the curve $y = x^3 - x$ at three distinct points. Find the complete set of values of m .

Answer: E

$$x^3 - x - mx = x(x^2 - (1 + m)) = 0: \text{ three distinct roots iff } 1 + m > 0, \text{ i.e. } m > -1.$$

Question 14 DIF-02 · Differentiation · spec MM6.3

Find the coordinates of the stationary points of $y = x^3 - 3x^2 - 9x + 5$ and state their nature.

Answer: C

$$y' = 3x^2 - 6x - 9 = 3(x - 3)(x + 1), y'' = 6x - 6: \text{ negative at } x = -1 \text{ (max, } y = -1 - 3 + 9 + 5 = 10), \text{ positive at } x = 3 \text{ (min, } y = 27 - 27 - 27 + 5 = -22).$$

Question 15 TRI-14 · Triangles, radians and sectors · spec MM4.1

A vertical pole BT stands on horizontal ground. From a point A on the ground the angle of elevation of T is 30° ; from a point C on the ground, 20 m closer to the pole than A and in line with A and B , the angle of elevation is 60° . Find the height of the pole.

Answer: D

Let the height be h . $AB = h\sqrt{3}$ and $CB = \frac{h}{\sqrt{3}}$. $AB - CB = 20 \Rightarrow h\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{2h}{\sqrt{3}} = 20 \Rightarrow h = 10\sqrt{3}$.

Question 16 POL-16 · Polynomials, factor and remainder theorems · spec MM1.6

For which value of k does $x^2 - x + k$ divide exactly into $x^3 - 2x^2 + 3x - 4$? (Assume such a value exists.)

Answer: F

Try $x^3 - 2x^2 + 3x - 4 = (x^2 - x + k)(x + m)$: comparing x^2 : $m - 1 = -2 \Rightarrow m = -1$. Then x : $k - m = k + 1 = 3 \Rightarrow k = 2$. Constant: $km = -2 \neq -4$. No value of k works.

Question 17 SEQ-19 · Sequences and series · spec MM2.3

The first three terms of a geometric sequence are x , $x + 6$ and $x + 15$. Find the fourth term.

Answer: A

Geometric: $(x + 6)^2 = x(x + 15) \Rightarrow x^2 + 12x + 36 = x^2 + 15x \Rightarrow x = 12$. The terms are 12, 18, 27 with ratio $\frac{3}{2}$, so the fourth term is $27 \times \frac{3}{2} = \frac{81}{2}$.

Question 18 QUA-22 · Quadratic functions · spec MM1.3

The equation $x^2 + ax + b = 0$ has roots 3 and -5 . The equation $x^2 + bx + a = 0$ has roots p and q . Find $p^2 + q^2$.

Answer: D

Sum of roots $3 + (-5) = -a$, so $a = 2$; product $-15 = b$. Then $x^2 - 15x + 2 = 0$ has $p + q = 15$ and $pq = 2$, so $p^2 + q^2 = 15^2 - 2(2) = 221$.

Question 19 IND-07 · Indices and surds · spec MM1.1

Which of the following is equal to $\frac{x^2\sqrt{x}}{\sqrt[3]{x}}$ for $x > 0$?

Answer: F

The index is $2 + \frac{1}{2} - \frac{1}{3} = \frac{12+3-2}{6} = \frac{13}{6}$.

Question 20 TRI-08 · Triangles, radians and sectors · spec MM4.2

The perimeter of a sector of radius r and angle θ radians is 20, and its area is 16. Find the possible values of r .

Answer: D

$2r + r\theta = 20$ and $\frac{1}{2}r^2\theta = 16$. From the first $r\theta = 20 - 2r$; then $\frac{1}{2}r(20 - 2r) = 16 \Rightarrow r^2 - 10r + 16 = 0 \Rightarrow r = 2$ or 8 .